



COLLABORATIVE SKILL EXCHANGE PLATFORM

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Abstract

- Capacity Connect is a web-based skill exchange platform that enables students to teach skills they know and learn skills they want through peer-to-peer interaction.
- It connects learners and mentors through skill matching, user profiles and exchange requests, promoting collaborative learning.

Problem Statement & Objectives

- Problem:** Students often struggle to find peers with the skills they want to learn, while opportunities to share their own expertise remain scattered and unorganized.
- Objective 1:** Provide a centralized platform for students to showcase their skills and learning interests.
- Objective 2:** Enable students to discover suitable learning and teaching partners.
- Objective 3:** Simplify skill exchange requests and communication between students.
- Scope:** Student registration, authentication, skill management, skill matching, exchange requests and user profiles.

Existing vs Proposed System

Feature	Existing System	Proposed System
Skill discovery	Manual searching	Centralized skill listings
Learning partners	Informal connections	Skill-based matching
Skill profiles	Scattered information	Organized user profiles
Exchange requests	Manual coordination	Digital request management
Learning opportunities	Limited visibility	Accessible peer-to-peer exchange

Feasibility Study

- Technical:** Uses React.js, Node.js, Express.js and MySQL, supported by widely available development tools.
- Economic:** Built using open-source technologies, reducing software and development costs.
- Operational:** Provides a web-based interface for students to manage skills and learning requests.
- Schedule:** Can be developed and tested in phases within the semester timeline.

Results & Testing

- User registration and login interfaces provide access to the platform.
- Skill listings and user profiles support skill discovery.

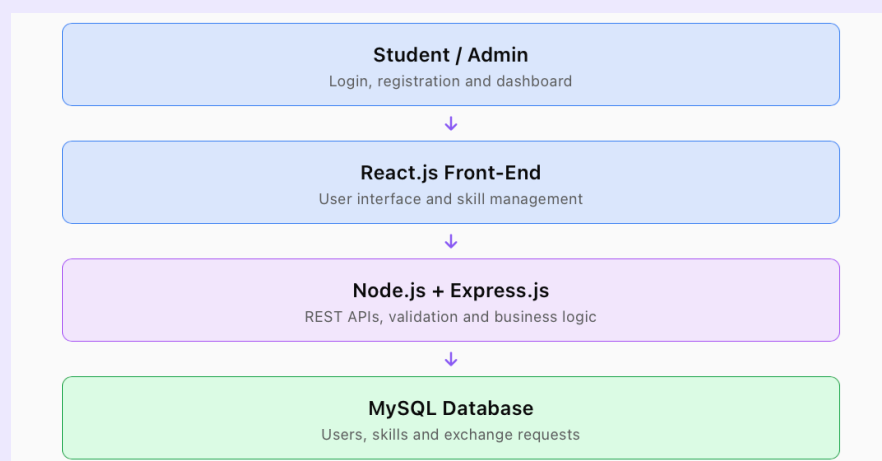
Conclusion

- Capacity Connect provides a centralized platform for peer-to-peer skill exchange among students.
- It aims to simplify skill discovery, encourage collaborative learning and make knowledge sharing more accessible.

Tools & Technologies

- Front-End: [React / HTML-CSS-JS]
- Back-End: [Node.js / Spring Boot / Django] + REST APIs
- Database: [MySQL / PostgreSQL / MongoDB]
- Tools: [Git & GitHub, Postman, VS Code, Draw.io]

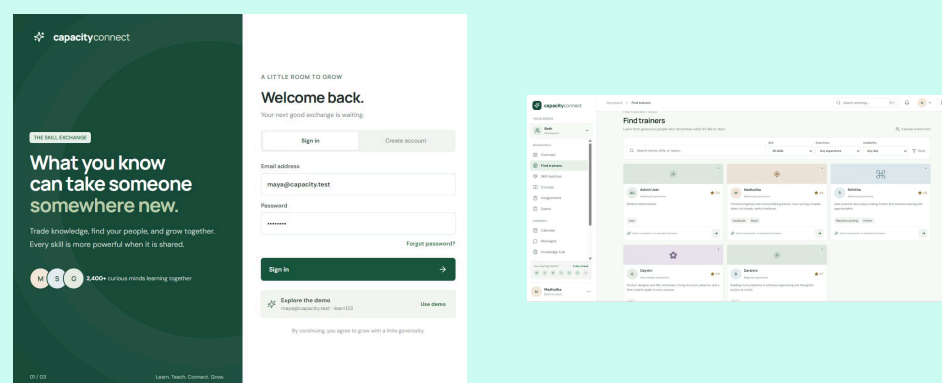
System Architecture



ER Diagram & Database Design



Implementation Screenshots



Future Scope

- Real-time communication:** Add integrated chat and notifications for learners and mentors.
- AI-based skill matching:** Recommend learning partners based on skills, interests and learning goals.



QR: GitHub